

# Planning, Learning and Intelligent Decision Making

## Homework 3 - POMDPs

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### Exercise 1.

The POMDP presented in the homework can be described by  $(\mathcal{X}, \mathcal{A}, \mathcal{Z}, \{P_a\}, \{O_a\}, c)$ , that will be defined in question 1.a. and 1.b.

(a)

- State space,  $\mathcal{X} = \{1, 2a, 2b, 3, 4\}$ ;
- Action space,  $\mathcal{A} = \{a, b, c\}$ ;
- Observation space,  $\mathcal{Z} = \{1, 2, 3, 4\}$ .

(b)

- $\{P_a\}$ :

$$P_a = \begin{matrix} & \begin{matrix} 1 & 2a & 2b & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2a \\ 2b \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 0.5 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \end{matrix}, \quad P_b = P_c = \begin{matrix} & \begin{matrix} 1 & 2a & 2b & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2a \\ 2b \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 0.5 & 0.5 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

- $\{O_a\}$ :

$$O_a = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2a \\ 2b \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \end{matrix}, \quad O_b = O_c = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2a \\ 2b \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

- Cost function,  $c$ :

$$c = \begin{matrix} & \begin{matrix} a & b & c \end{matrix} \\ \begin{matrix} 1 \\ 2a \\ 2b \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 \\ 0.5 & 1 & 0 \\ 0.5 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

**Note:** Actions b and c have the same transition and observation probabilities matrices.

(c)

At instant  $t$ , the agent's belief is:

$$b_t = [0.0 \quad 0.5 \quad 0.5 \quad 0.0 \quad 0.0]$$

Therefore, if the agent selects an action at time-step  $t$ , the updated agent's belief must consider the "prior" belief and the transition probability of the action selected.

- selects action  $a$  at time-step  $t$ :

$$b_{t+1} = b_t P_a = \begin{bmatrix} 0.0 & 0.5 & 0.5 & 0.0 & 0.0 \end{bmatrix} \begin{bmatrix} 0 & 0.5 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0.0 & 0.0 & 0.0 & 0.5 & 0.5 \end{bmatrix}$$

- selects action  $b$  at time-step  $t$ :

$$b_{t+1} = b_t P_b = \begin{bmatrix} 0.0 & 0.5 & 0.5 & 0.0 & 0.0 \end{bmatrix} \begin{bmatrix} 0 & 0.5 & 0.5 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1.0 & 0.0 & 0.0 & 0.0 & 0.0 \end{bmatrix}$$

- selects action  $c$  at time-step  $t$ :

$$b_{t+1} = b_t P_c = \begin{bmatrix} 0.0 & 0.5 & 0.5 & 0.0 & 0.0 \end{bmatrix} \begin{bmatrix} 0 & 0.5 & 0.5 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1.0 & 0.0 & 0.0 & 0.0 & 0.0 \end{bmatrix}$$